Analysis of The Morris Worm Incident of 1988 and The Lasting Effects on the Future

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Abstract

This paper will explore the foundation and motive behind the Morris Worm of 1988. Created by Robert Tappan Morris in 1988, a Cornell University graduate student. This paper will delve into the ethical and unethical aspects of its creation and use. It will also explain the methods that the Morris Worm used to spread and cause carnage across the nation. Lastly, it will address the legal side of it, along with the creation of CERT, and how it helped form the cyberworld as we know it today.

On November 2, 1988, a malicious program known as the Morris Worm was released onto the Internet, becoming one of the first major cyberattacks in history and earning it widespread recognition. Robert Tappan Morris, a 23-year-old Cornell University graduate student, created the Morris Worm. The program began to spread at remarkable speeds, infecting network systems, prestigious colleges, and public and private research centers. Robert Morris initially developed the program as a harmless experiment to see how far computer worms could propagate through the Internet (Funnell, 2023). The entirety of the event highlighted the numerous vulnerabilities present in the early Internet, as well as the potential implications of future cybersecurity laws and policies.

The Morris Worm worked by targeting particular computers running the Unix operating System. Once the Morris worm was connected to the computer, it attempted to propagate by copying itself to other computers. Its primary source of attack was through multiple vectors featuring a bug in the “finger” program, which identified network users. Along with exploiting the Internet’s electronic mail system, and guessing weak passwords. In addition to all these features, its primary design was to remain hidden until it was time to infect.

It was designed to check if the machine had already been affected by a Morris Worm program, and if it were, it would re-infect it 1 in 7 times. That “ensured that the user could not avoid a Morris worm infection by creating a fake Morris Worm”(Radware) to pretend that their machine was already infected. There was also the risk that a user's machine would become infected repeatedly, resulting in malfunctions. Within 24 hours, an estimated 6,000 of approximately 60,000 computers were under attack from the Morris Worm. The Morris Worm was able to spread so quickly due to its ability to replicate without needing a software host;

instead, it only required itself to function. It wasn't designed to destroy files or damage them, but rather to slow and delay computers.

Multiple research centers, government facilities, and even universities were in an uproar over the virus. The entire world of networking worked overtime just to figure out how to remove it. In some organizations, IT professionals decided to wipe their systems clean, while others disconnected from the internet for a week or less. The damages surrounding the virus initially totaled around $100,000 and increased into the millions, with some additional damages remaining unaccounted for. It became more urgent to find the culprit of the program rather than how to stop it, and computer experts were soon to rejoice.

Shortly after the Morris Worm was launched, a distressed programmer told two of his friends that he was the one who had launched the worm and lost control of it. He then asked one of his friends to pass on an anonymous message on his behalf, stating that he was sorry and providing a guide on how to remove the program. To his dismay, the message was delayed due to the worm spreading like wildfire. His other friend, however, contacted The New York Times, stating that he knew who built the program and its intended use. Unfortunately, his friend ratted Robert Morris out, and soon the FBI launched an investigation.

Immediately, they discovered that Morris was behind the Worm virus and began to interview and search his belongings. In his computer files, they found plenty of incriminating evidence that had broken federal law. In 1986, Congress passed the CFAA, or Computer Fraud and Abuse Act, which prohibits unauthorized access to a computer. This, in turn, made it illegal to break into systems, use malicious code, and damage data in other systems. Prosecutors used this to indict Morris in 1989, and the following year, he was found guilty, making him the first person ever to be convicted under the CFAA.

Morris, to his luck, was spared jail time but instead received a fine, probation, and a direct order to complete 400 hours of community service. This entire event not only showcased the vulnerability of computers and software but also made a call to action. Considering how spontaneous the virus was, the coordination was also all over the place, which played a massive role in maintaining systems and stopping the virus. At that moment, it revealed how unprepared the networking world was for a cyberattack, along with the lack of necessary protocols.

This large-scale cyberattack highlighted the need for a cyberdefense team known as the Computer Emergency Response Team, or CERT. DARPA, also known as the U.S. Department of Defense's Defense Advanced Research Projects Agency, provided the necessary funding for the CERT program. Before CERT was formed, universities, research centers, and government facilities had to use their own discretion and skills to manage a wide-scale cyberattack. Ever since the Morris Worm takeover, CERT has served as a hub of information and problem-solving regarding all computer-related matters, along with a team that can come together to provide solutions for problems. Although CERT doesn't deal with code directly, it does offer solutions or guidance on the best way to approach a cyberattack.

Even today, CERT has become a model for all forms of cyber protocols dating back to the Morris Worm. Pushing past the events of the Morris Worm raised serious ethical questions regarding responsibility for such a wide-scale event. Robert Morris claimed that he had no malicious intent, but at the same time, it was debated that he should have expected how releasing an untested program into live systems would go. Morris Worm demonstrated how little research can be dangerous in a world of computers, ethically and carelessly. On the other hand, had this

event not occurred, systems would not be where they are today, with continuous upgrades in security and thousands of methods to mitigate cyberattacks.

It also highlighted the importance of accountability in the field, reiterating that good intentions do not absolve one of responsibility when real harm can occur. As mentioned before, the Morris Worm, although malicious code, did some good in demonstrating how to improve network security. In that day and age, many old systems relied heavily on weak passcodes, which could easily be guessed by brute force attacks. There were also software vulnerabilities, such as Unix sendmail and finger. While these weren't over-the-top hacking methods, if not correctly patched, they could cause some damage.

Overall, a stronger policy on passwords, around-the-clock updates, and even better authentication could have been factors in reducing the damage caused by the Morris Worm. One key takeaway could be the lack of communication during the cyberattack. Systems and communication were down, along with email. While this doesn't seem like an issue, it is. Every form of communication was either down or prolonged due to the Morris Worm. When something as drastic as a cyberattack is in play, communication is a key feature that is needed, along with the lack of coordinated response due to the worm, which plays a massive role in delayed containment.

Today, the incident is widely discussed as both a major cyberattack and a catalyst for improved security, communication, and policies in the event of another major cyberattack. All in all, the Morris Worm really opened our eyes to the lack of cyber practices we had in place. Initially, the Morris Worm was considered a fun experiment, but it ultimately evolved into the first major cyberattack in history. This led to the creation of CERT, which was the first cyber defense designed to provide guidance and respond to major cyberattacks. During this time,

Congress passed the CFAA, or Computer Fraud and Abuse Act, whose task was to prohibit the unauthorized use of computers.

From an ethical perspective, it highlighted the importance of accountability and responsibility when dealing with something so dangerous. It also taught future coders to proceed with caution and know the risks should it ever get out of hand. In the long run, the Morris Worm was both a blessing and a curse in the cyber field. Mainly teaching us how our systems lacked pure security and were vulnerable. It also kick-started the development of stronger security measures, specific laws, and an overall sense of awareness about the dangers of the cyberworld.

Finally, the Morris Worm, in a way, paved the way for hackers and criminals to create malicious code that could compromise and potentially destroy all our network systems. On the other hand, the government and cybersecurity pros were able to build on the skills from all those years ago. Thus producing stronger protection, better policies, and a more stable way of communication between people. Ultimately, the Morris Worm laid the groundwork for future advancements.

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